

DELFT UNIVERSITY OF TECHNOLOGY

Faculty of Electrical Engineering, Mathematics and Computer Science



Final assignment
Microsystems Design and Modelling (ET4260)
Part 1/3

Instructions. Document your calculations properly with derivations and screenshots when needed. Latex-generated PDF files are accepted.

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Deadline for final report submission: 24th June 2026.

Corrections made after the first submission are marked in [blue](#).

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Optical phase shifter for quantum applications

The MEMS photonic phase shifter studied in this report (Fig. 1) modulates the optical phase by changing the gap g between a suspended silicon mass and an adjacent waveguide. The mass is suspended by four fixed-guided beams and displaced by a comb-drive actuator biased with V_d . With the gap varying as $g = g_0 + x_m$, the additional phase shift is

$$\Delta\phi(g) = \frac{2\pi L}{\lambda} (n_{\text{eff}}(g_0) - n_{\text{eff}}(g)), \quad n_{\text{eff}}(g) = (\eta_{\text{eff},0} - \eta_{\text{eff},\infty}) e^{-bg} + \eta_{\text{eff},\infty}. \quad (1)$$

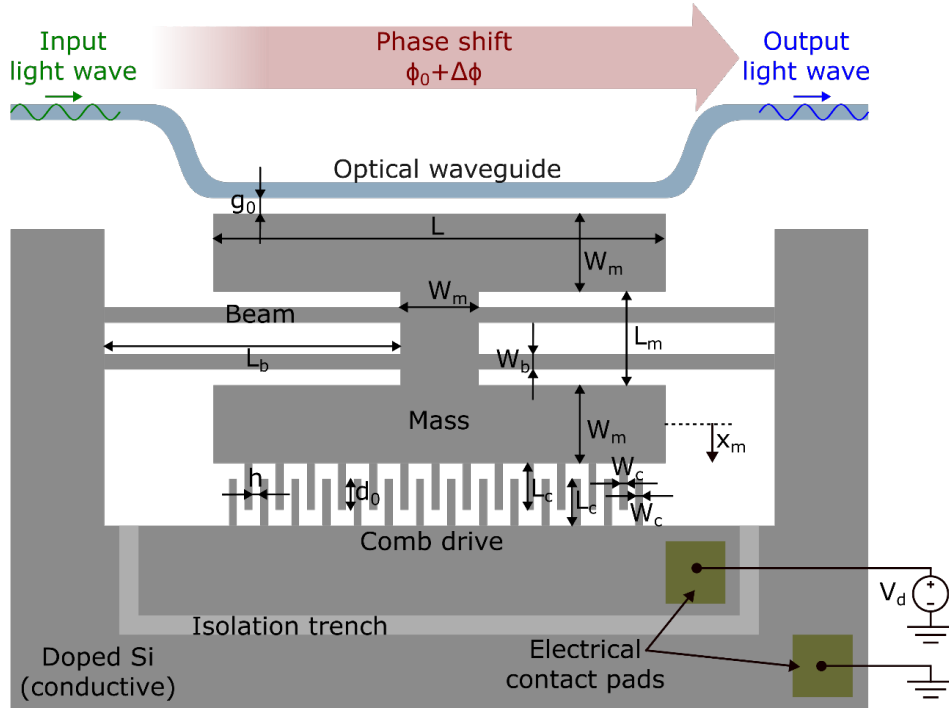


Figure 1: Schematic of the MEMS optical phase shifter (top view, not to scale). The waveguide sits a gap g_0 above the modulation section. The central mass is suspended by four fixed-guided beams of length L_b and width W_b , anchored on either side. The comb drive at the bottom generates the actuating force F_e .

Table 1: Design parameters used throughout this report.

Symbol	Value	Description
L	100 μm	length of modulation section
g_0	200 nm	modulation gap at rest
L_b	75 μm	beam length
W_b	5 μm	beam width
L_m	40 μm	mass central section length
W_m	30 μm	mass section width
h	1 μm	comb-drive gap
d_0	19 μm	comb overlap at rest
L_c	20 μm	comb-finger length
W_c	1 μm	comb-finger width
t	200 nm	thickness of suspended layer
N	40	number of comb-drive gaps
E_{Si}	170 GPa	Young's modulus of silicon
ρ_{Si}	2330 kg/m ³	density of silicon
λ	1550 nm	optical wavelength
b	0.3/ μm	refractive-index decay constant
$\eta_{\text{eff},0}$	2.6	n_{eff} at $g = 0$
$\eta_{\text{eff},\infty}$	2.4	n_{eff} as $g \rightarrow \infty$
c_{air}	16.15 N s/m ³	air damping coefficient
Q_{clamping}	500	clamping quality factor

1 Mechanical model and dynamics

Q1 – Lumped-element equation of motion

In vacuum and without damping, the suspended mass behaves as a single spring–mass with input force F_e and output displacement x_m :

$$m \ddot{x}_m + k x_m = F_e. \quad (2)$$

Once air damping and clamping losses are included (Q4) the equation extends to

$$m \ddot{x}_m + b \dot{x}_m + k x_m = F_e, \quad (3)$$

which is the classical mass–spring–damper system, equivalent to a series RLC circuit (Fig. 2).

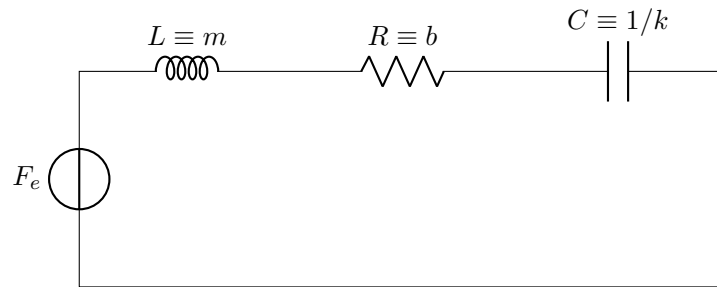


Figure 2: Series-RLC equivalent circuit of the lumped mass–spring–damper system. Effort $\leftrightarrow$ voltage, flow $\leftrightarrow$ current.

Q2 – Fundamental resonance frequency

Stiffness. Each of the four beams is clamped at its anchor and connected to the rigid mass at the other end. By the symmetry of the upper/lower beam pair, the mass-side end translates

without rotation – a *fixed-guided* boundary condition. Each beam therefore contributes

$$k_b = \frac{12 E_{Si} I}{L_b^3}, \quad I = \frac{t W_b^3}{12}, \quad (4)$$

where the in-plane width W_b is the dimension along the bending direction. With four beams in parallel,

$$k = 4k_b = \frac{4 E_{Si} t W_b^3}{L_b^3} = 40.30 \text{ N/m}. \quad (5)$$

Effective mass. Neglecting beam mass, the moving body comprises the modulation strip, the central mass and the comb fingers:

$$m = \rho_{Si} t (L W_m + L_m W_m + N L_c W_c) = 2.33 \times 10^{-12} \text{ kg}. \quad (6)$$

Resonance.

$$\omega_0 = \sqrt{\frac{k}{m}} = 4.16 \times 10^6 \text{ rad/s}, \quad f_0 = \frac{\omega_0}{2\pi} \approx 662 \text{ kHz}. \quad (7)$$

Q3 – Static deflection due to gravity

With gravity acting along the direction of motion, static balance gives

$$x_g = \frac{m g}{k} = \frac{2.33 \times 10^{-12} \text{ kg} \cdot 9.81 \text{ m/s}^2}{40.30 \text{ N/m}} \approx 0.57 \text{ pm}. \quad (8)$$

This is six orders of magnitude smaller than g_0 . Gravity is irrelevant at this scale; the comb-drive force dominates by a wide margin.

2 Damping, settling time and shock tolerance

Q4 – Settling time with combined damping

Independent loss channels add in parallel: damping coefficients sum, quality factors combine as reciprocals.

Air damping. The top-side area of the moving structure is

$$A_m = L W_m + L_m W_m + N L_c W_c = 5.00 \times 10^{-9} \text{ m}^2, \quad (9)$$

giving an air damping coefficient

$$b_{\text{air}} = c_{\text{air}} A_m = 8.08 \times 10^{-8} \text{ N s/m}, \quad Q_{\text{air}} = \frac{\sqrt{km}}{b_{\text{air}}} = 120. \quad (10)$$

Clamping loss. From the given $Q_{\text{clamping}} = 500$,

$$b_{\text{clamp}} = \frac{\sqrt{km}}{Q_{\text{clamping}}} = 1.94 \times 10^{-8} \text{ N s/m}. \quad (11)$$

Combined damping.

$$b = b_{\text{air}} + b_{\text{clamp}} = 1.00 \times 10^{-7} \text{ N s/m}, \quad \frac{1}{Q} = \frac{1}{Q_{\text{air}}} + \frac{1}{Q_{\text{clamping}}} \Rightarrow Q \approx 96.8. \quad (12)$$

Air damping dominates ($Q_{\text{air}} = 120 \ll Q_{\text{clamping}} = 500$), so Q is essentially set by the surrounding gas.

Settling time. With $\zeta = b/(2\sqrt{km}) = 1/(2Q) \approx 5.2 \times 10^{-3}$ the system is strongly underdamped. Its envelope decays as $e^{-\zeta\omega_0 t}$, giving a $\pm 5\%$ settling time of

$$t_s \approx \frac{-\ln(0.05)}{\zeta\omega_0} = \frac{3.00}{\zeta\omega_0} \approx 139 \mu\text{s}. \quad (13)$$

Figure 3 shows the simulated response for a 1 nN step input.

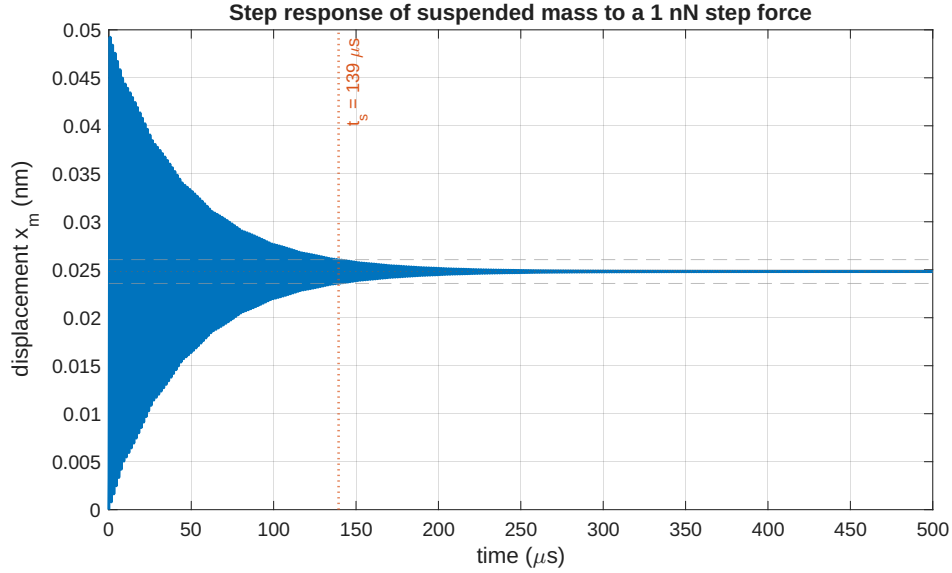


Figure 3: Step response to a 1 nN input force. The dashed lines mark the $\pm 5\%$ band around the steady-state value F_0/k . Settling occurs after roughly $t_s \approx 139 \mu\text{s}$.

Q5 – Maximum tolerable harmonic acceleration

A harmonic base acceleration $a(t) = A \cos(\omega t)$ acts on the suspended mass as an inertial force $-ma$. With $V_d = 0$ (no electrostatic force), the equation of motion in the frequency domain is

$$(k - m\omega^2 + j\omega b) X(j\omega) = -m A(j\omega), \quad (14)$$

so the magnitude of the displacement is

$$|X(j\omega)| = \frac{m |A(j\omega)|}{\sqrt{(k - m\omega^2)^2 + (\omega b)^2}}. \quad (15)$$

The mass touches the waveguide when $|X| = g_0$. The maximum tolerable acceleration as a function of frequency is therefore

$$A_{\max}(\omega) = \frac{g_0}{m} \sqrt{(k - m\omega^2)^2 + (\omega b)^2}. \quad (16)$$

DC limit ($\omega \rightarrow 0$).

$$A_{\max, \text{DC}} = \frac{g_0 k}{m} = g_0 \omega_0^2 \approx 3.46 \times 10^6 \text{ m/s}^2 \approx 3.5 \times 10^5 g. \quad (17)$$

Worst-case AC frequency ($\omega = \omega_0$). At resonance the square root reduces to $\omega_0 b$, so

$$A_{\max, \text{res}} = \frac{g_0 \omega_0 b}{m} = \frac{g_0 \omega_0^2}{Q} \approx 3.6 \times 10^4 \text{ m/s}^2 \approx 3.6 \times 10^3 g. \quad (18)$$

Resonance reduces shock tolerance by a factor Q . The DC value ($\sim 10^5 g$) is enormous, but at f_0 the device is roughly $100\times$ more vulnerable. Fig. 4 plots $A_{\max}(f)$.

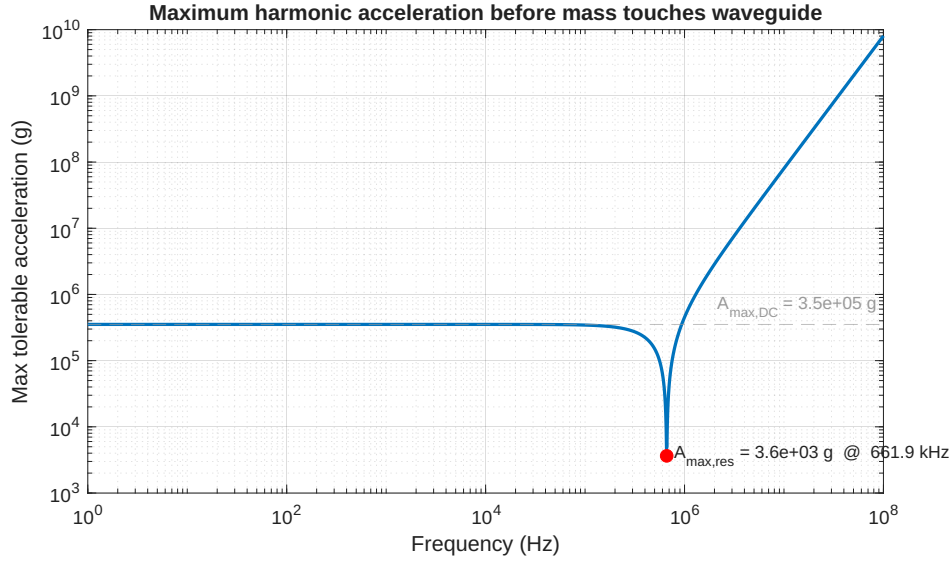


Figure 4: Maximum tolerable acceleration before the suspended mass contacts the waveguide, computed from Eq. 16. The minimum lies at $f_0 \approx 662$ kHz, where the device is most sensitive to mechanical excitation.

3 Electrostatic transduction and sensitivity

Q6 – Static phase shift versus DC voltage

Considering only the perpendicular field component, the comb-drive capacitance varies linearly with the engagement length $d_0 + x_m$:

$$C(x_m) = N \varepsilon_0 \frac{t(d_0 + x_m)}{h}, \quad \frac{dC}{dx_m} = \frac{N \varepsilon_0 t}{h} = 7.08 \times 10^{-11} \text{ F/m}. \quad (19)$$

The electrostatic force on the mass therefore depends only on voltage:

$$F_e = \frac{1}{2} V_d^2 \frac{dC}{dx_m} = \frac{1}{2} V_d^2 \frac{N \varepsilon_0 t}{h}. \quad (20)$$

Static balance with the spring force ($F_e = kx_m$) yields the displacement

$$x_m(V_d) = \frac{V_d^2 N \varepsilon_0 t}{2kh}, \quad g(V_d) = g_0 + x_m(V_d). \quad (21)$$

Substituting into Eq. 1 gives the phase shift plotted in Fig. 5. It is symmetric in V_d and quadratic. At $V_d = \pm 50$ V the gap only changes by $x_m \approx 2.2$ nm (1% of g_0), so $|\Delta\phi|$ stays around 50 mrad. Reaching π radians of phase shift would need a much higher bias.

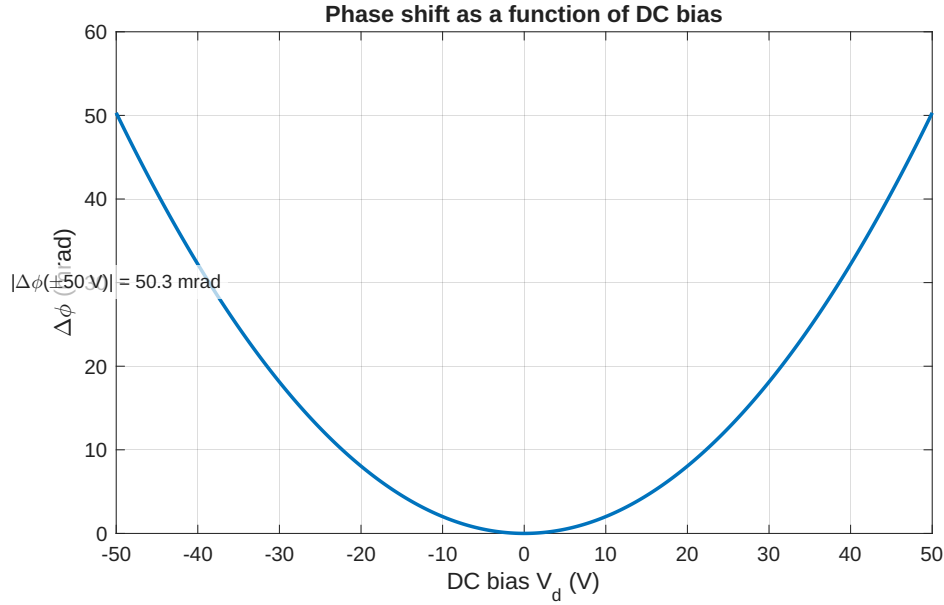


Figure 5: Static phase shift $\Delta\phi$ as a function of the DC bias $V_d \in [-50, +50]$ V. The shift is symmetric and quadratic in V_d , in line with Eq. 20.

Q7 – Pull-in instability of the comb drive

Case A – perpendicular field only. The overlap force in Eq. 20 does not depend on x_m , so $dF_e/dx_m = 0 < k$ for any V_d . The single equilibrium $x_m = \frac{1}{2}V_d^2(dC/dx_m)/k$ is always stable. *No pull-in.* This is why comb drives are preferred over parallel-plate actuators when long strokes are needed.

Case B – including finger-tip parallel-plate capacitance. When the mass moves into the comb, the gap between each finger tip and the opposing anchor shrinks from $g_{t,0} = L_c - d_0 = 1\text{ }\mu\text{m}$ to $g_t = g_{t,0} - x_m$. The additional capacitance and force per finger pair are

$$C_{\text{tip}}(x_m) = N \varepsilon_0 \frac{W_c t}{g_{t,0} - x_m}, \quad F_{\text{tip}}(x_m) = \frac{1}{2} V_d^2 \frac{N \varepsilon_0 W_c t}{(g_{t,0} - x_m)^2}. \quad (22)$$

Total force balance:

$$kx_m = \frac{1}{2} V_d^2 \left[\frac{N \varepsilon_0 t}{h} + \frac{N \varepsilon_0 W_c t}{(g_{t,0} - x_m)^2} \right]. \quad (23)$$

Pull-in occurs when the small-signal stability condition $dF/dx_m = k$ is reached:

$$V_d^2 \frac{N \varepsilon_0 W_c t}{(g_{t,0} - x_m)^3} = k. \quad (24)$$

Eliminating V_d^2 between the two equations gives an implicit relation for the pull-in gap $u \equiv g_{t,0} - x_{\text{pi}}$:

$$g_{t,0} = \frac{3u}{2} + \frac{u^3}{2hW_c}. \quad (25)$$

Numerically, $u \approx 0.60\text{ }\mu\text{m}$ (so $x_{\text{pi}} \approx 0.40\text{ }\mu\text{m}$), giving the pull-in voltage

$$V_{\text{PI}} = \sqrt{\frac{k u^3}{N \varepsilon_0 W_c t}} \approx 347\text{ V}. \quad (26)$$

The mass-to-waveguide contact would only happen at $V_d \approx \sqrt{2kg_0/(dC/dx_m)} \approx 477\text{ V}$, so finger-tip pull-in is what limits operation. As long as the bias stays well below $\sim 350\text{ V}$, Case A is a good model and the device behaves linearly.

4 Small-signal modulation response

Q8 – Transfer function from V_{AC} to $\Delta\phi$

Linearised force. With $V_d(t) = V_{DC} + V_{AC} \sin(2\pi f_m t)$ and $V_{AC} \ll V_{DC}$, the comb-drive force expands to first order as

$$F_e = \frac{1}{2} (V_{DC} + V_{AC})^2 \frac{dC}{dx_m} = \underbrace{\frac{1}{2} V_{DC}^2 \frac{dC}{dx_m}}_{F_{DC}} + \underbrace{V_{DC} V_{AC} \frac{dC}{dx_m}}_{f_{AC}(t)} + \mathcal{O}(V_{AC}^2). \quad (27)$$

The AC component drives the small-signal motion with electromechanical coupling

$$\eta = V_{DC} \frac{dC}{dx_m} = V_{DC} \frac{N \varepsilon_0 t}{h}. \quad (28)$$

The frequency response of the mass displacement is then

$$\frac{X(s)}{V_{AC}(s)} = \frac{\eta}{ms^2 + bs + k}, \quad (29)$$

a standard second-order low-pass with poles at the resonance frequency.

Phase-shift sensitivity. Linearising Eq. 1 around the DC operating gap $g_{op} = g_0 + x_m(V_{DC})$ via a first-order Taylor expansion,

$$\Delta\phi(g_{op} + \delta g) \approx \Delta\phi(g_{op}) + \underbrace{\left. \frac{d\Delta\phi}{dg} \right|_{g_{op}}}_{\equiv S_\phi} \delta g. \quad (30)$$

Using $g = g_0 + x_m$ so that $\delta g = \delta x_m$, and Eq. 1,

$$S_\phi = \frac{2\pi L}{\lambda} (\eta_{eff,0} - \eta_{eff,\infty}) b e^{-b g_{op}}. \quad (31)$$

Combined transfer function. Multiplying Eq. 29 by S_ϕ gives

$$H(s) = \frac{\Delta\phi(s)}{V_{AC}(s)} = \frac{S_\phi \eta}{ms^2 + bs + k}. \quad (32)$$

Performance metrics ($V_{DC} = 10 \text{ V}$).

$$|H|_{f \rightarrow 0} = \frac{S_\phi \eta}{k} \approx 0.40 \text{ mrad/V}, \quad (33)$$

$$|H|_{f=f_0} = \frac{S_\phi \eta Q}{k} \approx 39 \text{ mrad/V} = Q \cdot |H|_{f \rightarrow 0}, \quad (34)$$

$$\text{roll-off} = -40 \text{ dB/decade} \quad (\text{two-pole low-pass above } f_0). \quad (35)$$

Fig. 6 shows the Bode plot. The magnitude is flat below f_0 , peaks by $20 \log_{10} Q \approx 40 \text{ dB}$ at the lightly damped resonance, then rolls off at -40 dB/dec .

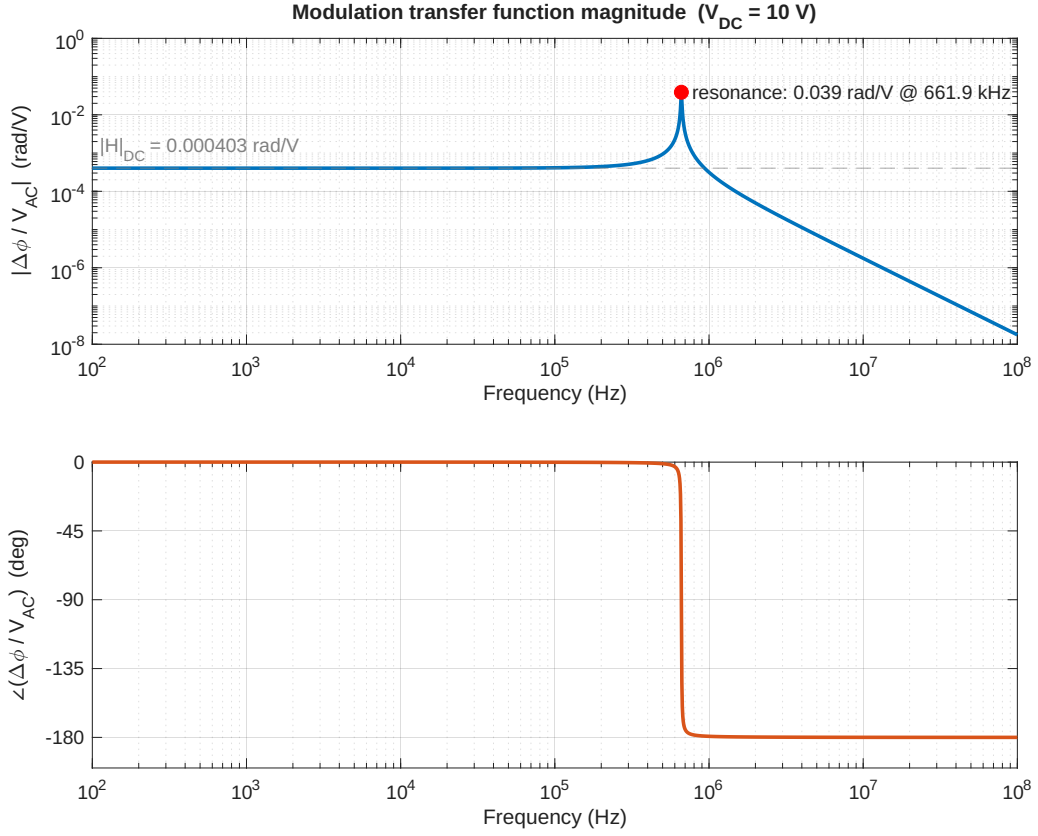


Figure 6: Bode diagram of the small-signal transfer function $\Delta\phi(s)/V_{AC}(s)$ at $V_{DC} = 10$ V. The peak at $f_0 \approx 662$ kHz exceeds the low-frequency value by a factor of $Q \approx 97$.

Low-frequency sensitivity vs DC bias. From Eqs. 28 and 32, $|H|_{f \rightarrow 0} = S_\phi V_{DC} (N\epsilon_0 t/h)/k \propto V_{DC}$ to first order. Sensitivity scales linearly with V_{DC} because η does. Fig. 7 shows the response over $V_{DC} \in [0, 5]$ kV, with the linear small-signal prediction and the result that includes the operating-point shift in S_ϕ . Two practical limits are also marked: finger-tip pull-in at $V_{PI} \approx 347$ V and mass-waveguide contact at ≈ 477 V. Above ~ 350 V the device snaps down, so most of the plotted range is unreachable in practice – it is shown only to illustrate the trend.

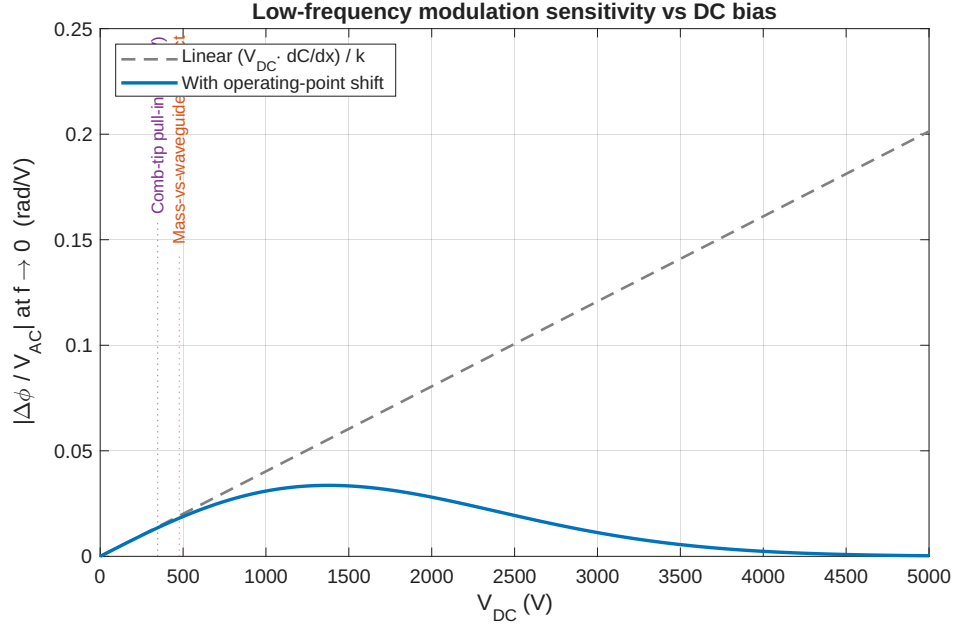


Figure 7: Low-frequency modulation sensitivity as a function of V_{DC} . The dashed grey line is the strict linear V_{DC} dependence; the solid line includes the (small) effect of the operating-point shift on S_ϕ . Practical operation is bounded by comb-tip pull-in at ~ 347 V.

5 Geometric scaling

Q9 – All dimensions multiplied by 1000

Let every linear dimension grow by $\alpha = 10^3$. Each derived quantity picks up the corresponding power of α from its formula. Results are collected in Table 2; the reasoning follows.

Table 2: Scaling of device characteristics for $\alpha = 10^3$.

Quantity	Power of α	Scale factor
Stiffness k	α^1	10^3
Effective mass m	α^3	10^9
Resonance $\omega_0 = \sqrt{k/m}$	α^{-1}	10^{-3}
Gravity displacement $x_g = mg/k$	α^2	10^6
Max AC accel. $A_{\max, \text{res}} = g_0 \omega_0^2 / Q$	α^{-1}	10^{-3}
Comb coupling $\eta = V_{DC} N \varepsilon_0 t / h$	α^0	1

Stiffness ($\propto \alpha$). $k = 4 E_{Si} t W_b^3 / L_b^3$ contains four lengths in the numerator (t, W_b^3) and three in the denominator (L_b^3), so $k \propto \alpha^{4-3} = \alpha$.

Mass ($\propto \alpha^3$). $m = \rho \cdot V_{\text{moving}}$ and volume scales as α^3 .

Resonance ($\propto \alpha^{-1}$). $\omega_0 = \sqrt{k/m} \propto \sqrt{\alpha/\alpha^3} = \alpha^{-1}$. The macroscale device would resonate at ~ 660 Hz instead of 660 kHz.

Gravity displacement ($\propto \alpha^2$). $x_g = mg/k \propto \alpha^3/\alpha = \alpha^2$. The 0.57 pm deflection becomes ~ 0.6 mm. Gravity, irrelevant here, becomes a dominant load at the macroscale.

Max AC acceleration ($\propto \alpha^{-1}$). $A_{\max, \text{res}} = g_0 \omega_0^2 / Q \propto \alpha \cdot \alpha^{-2} = \alpha^{-1}$ (assuming Q unchanged). Shock tolerance drops from $\sim 3600 g$ to a few g .

Coupling constant (α^0). $\eta = V_{\text{DC}} N \varepsilon_0 t/h$. With N fixed and t, h scaling together, t/h is invariant and η stays the same at a given V_{DC} .

The pattern is the usual one for MEMS: shrinking the device makes it stiffer relative to its mass (faster), shock-tolerant, and immune to gravity. Going in the opposite direction undoes all three.